# Umair Javaid Manj

Auckland, NZ | umairjavaidmanj@gmail.com | +64 20 4483186 | Research page | ORCID

## Research Experience

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| **Doctoral Researcher** | **Auckland University of Technology**, Health & Artificial Intelligence, Auckland<br>• Developing an AI based biomarker of chronic low back pain from cortical responses to paraspinal stimulation.<br>• Extending fMRI mapping of the back (Cole et al., 2022; Dörig et al., 2026) into the EEG dimension, adding millisecond timing and oscillatory dynamics, and comparing painful with healthy spinal segments within the same person.<br>• Designing a case control EEG study combining source imaging, representational similarity analysis and subject level machine learning.<br>• Led the first pilot and identified an analysis artefact behind an apparent decoding effect. | Oct 2025 to Present |
| **Data & AI Governance Analyst** | **New Zealand College of Chiropractic**, Auckland<br>• Analyse EEG hyperscanning of chiropractor and patient: inter brain and within brain connectivity (wPLI, dPLI) and source level network analysis.<br>• Apply graph theory to resting state EEG networks in a randomised chiropractic trial of adults and children, with linear mixed models.<br>• Model longitudinal clinical outcomes (immune response, DASS-21, PROMIS-25) with mixed effects models in R.<br>• Manage private participant data and consent records. | Aug 2026 to Present |
| **Program Manager** | **National University of Sciences and Technology**, Islamabad<br>• Led a national AI training program. | Sep 2024 to Oct 2025 |
| **Instructor** | **AI Lounge**, Islamabad<br>• Supervised 18 student AI projects; awarded Best Instructor. | May 2023 to Jul 2024 |
| **Research Associate** | **COGNET Lab, SEECS**, National University of Sciences and Technology, Islamabad<br>• Researched the robustness of deep learning and computer vision models to adversarial inputs.<br>• Mentored junior research assistants (Research Assistant until Nov 2022). | Aug 2021 to Apr 2023 |

## Publications

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. (2026). Application of machine learning techniques in the measurement of ground reaction force (GRF) waveforms using inertial measurement unit (IMU) sensors: a systematic literature review. *Medical Engineering & Physics*. doi
- Rizvi, D. Q., Majeed, M., **Manj, U. J.**, & Haq, M. I. U. (2025). Sleep disorder prediction with PSO-optimized machine and deep learning models. *ICODT2*, pp. 1 to 6. doi
- Haq, M. I. U., Ahmad, Z., **Manj, U. J.**, Akhter, M. N., Arfeen, Z. A., & Qamar, S. D. (2026). Bio-inspired soft robotic gripper with variable stiffness using magneto-rheological control. *ICRAI*, pp. 1 to 6. doi
- Haq, M. I. U., **Manj, U. J.**, Paracha, S. Q., Imam, Y., Rizvi, D. Q., & Farooqi, A. R. (2026). Neuro-symbolic AI-driven secure communication framework for autonomous cyber-physical systems. *APAN 2026*, pp. 1 to 6. doi
- Akhter, M. N., **Manj, U. J.**, Ali, H., Niazi, I. K., & Taylor, D. Deep learning-based kinetics and center of pressure estimation during sit-to-walk transitions from a single sacrum-mounted IMU. *IEEE Transactions on Biomedical Engineering* (under review).

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. Predicting ground reaction force during sit to walk: towards kinetically instrumenting the Timed Up and Go test. *Medical & Biological Engineering & Computing* (under review).
- **Manj, U. J.**, Rizvi, S. D. Q., Niazi, I. K., Akhter, M. N., & Ali, H. Gated feature fusion and age confounding in plantar thermography for diabetes screening. *Biomedical Physics & Engineering Express* (under review).
- **Manj, U. J.**, Qamar, D., Rashid, M. A., Perwaiz, N., & Niazi, I. K. Speaker leakage and subject level accuracy in voice biomarkers for Parkinson's disease. *Machine Learning: Health* (under review).

## Education

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| **PhD** | **Auckland University of Technology**, Health & Artificial Intelligence, Auckland, New Zealand | Oct 2025 to Present |
| **MS** | **National University of Sciences and Technology**, Computational Science & Engineering, Islamabad | 2017 to 2021 |
| **BE** | **National University of Sciences and Technology**, Computer Engineering, Islamabad | 2012 to 2016 |

## Skills

**Neurophysiology & Signals:** EEG preprocessing (ICA, ASR, GEDAI), ERP and time frequency analysis, source localisation, functional connectivity (wPLI, dPLI) and graph theory, hyperscanning, IMU analysis, vibrotactile stimulation

**EEG Toolboxes:** MNE-Python, autoreject, EEGLAB (ICLabel, ROIconnect), FieldTrip, BrainNet Viewer

**Machine Learning:** PyTorch, scikit-learn, SHAP; subject level evaluation, leakage and confounding checks

**Statistics:** probability and sampling, confidence intervals, hypothesis testing, linear mixed models (lme4, lmerTest, emmeans), cluster permutation tests, power analysis (G*Power)

**Programming:** Python (NumPy, SciPy, pandas, NetworkX), R, MATLAB, C++

**Research Practice:** Participant data governance, systematic review, scientific writing

## Referees

**Dr. Nazia Perwaiz**, MSCA Industrial Postdoctoral Fellow, School of Computer Science, University of Galway, Ireland; Assistant Professor, SEECS, National University of Sciences and Technology. nazia.perwaiz@seecs.edu.pk

**Dr. Haider Ali**, Faculty of Design and Creative Technologies, Auckland University of Technology; AI Engineer, KPMG New Zealand. haider.ali@autuni.ac.nz